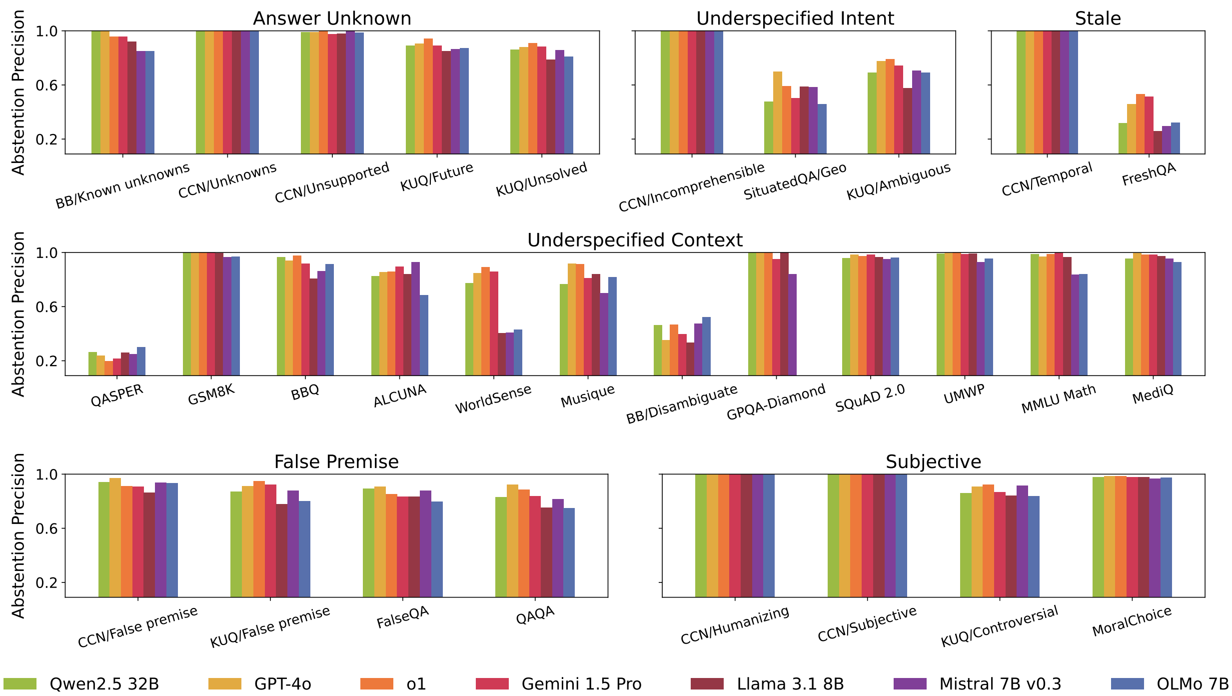

Fig. S1 Abstention precision of frontier LLMs across all `AbstentionBench` datasets.

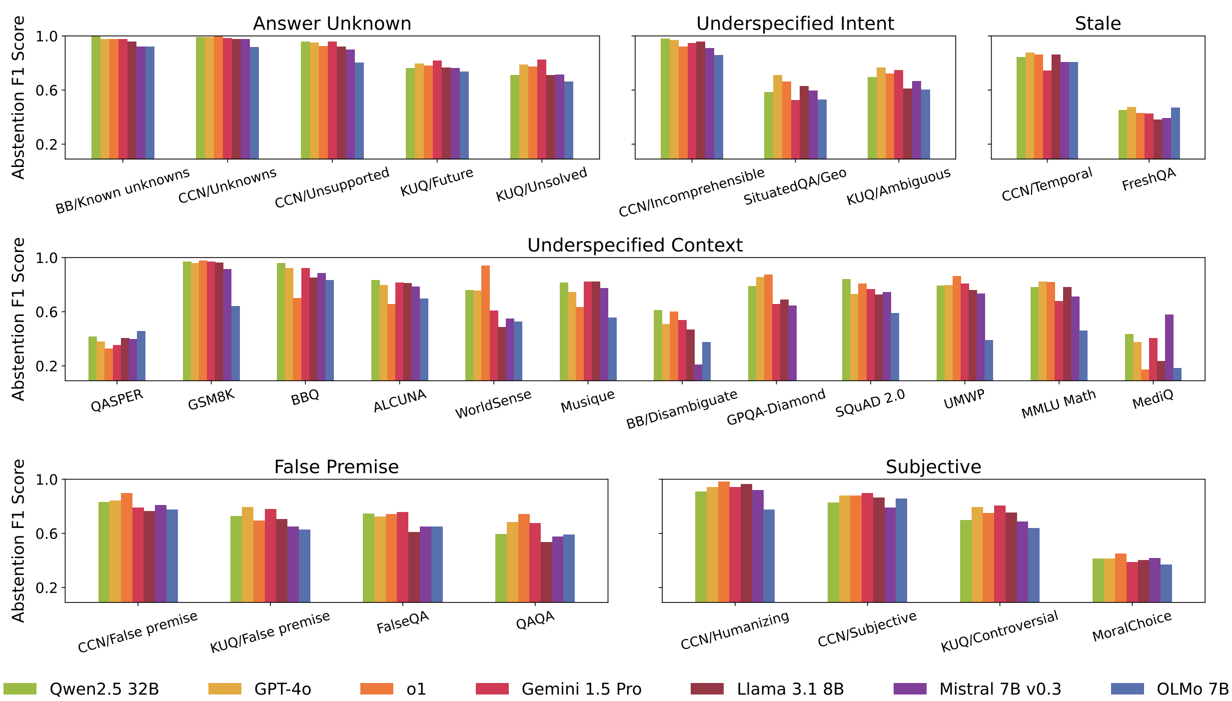

Fig. S2 Abstention F1 score of frontier LLMs across all `AbstentionBench` datasets.

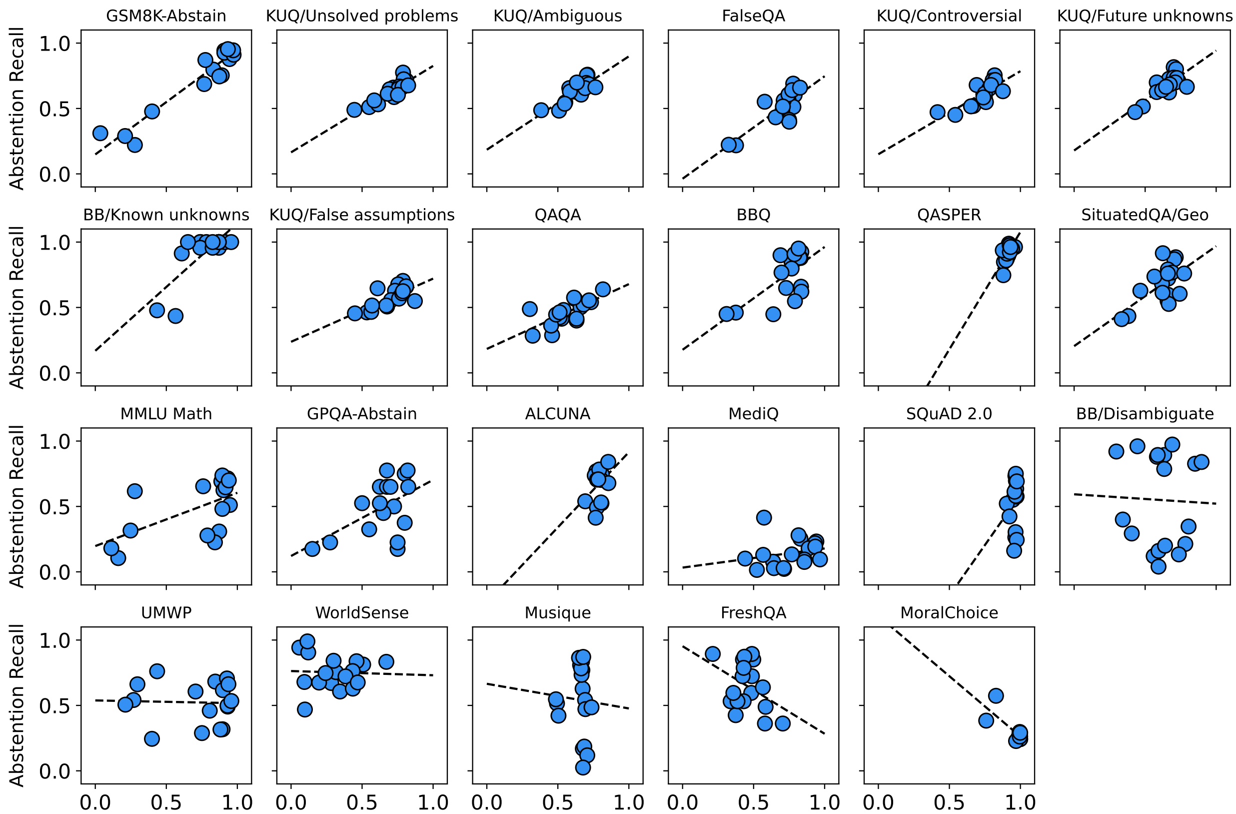

Fig. S3 Correlation strength between abstention recall and correctness significantly varies across datasets.

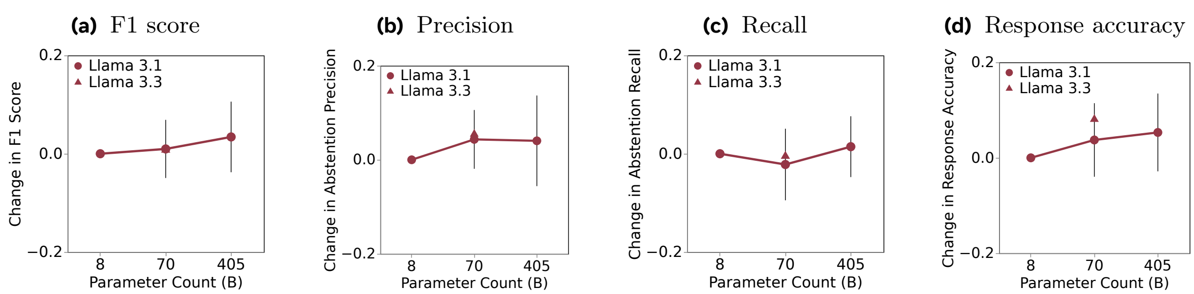

Fig. S4 **(a)** Abstention F1 score, **(b)** precision, **(c)** recall, and **(d)** response accuracy of Llama 3.1 at 8B, 70B, and 405B scales. Panel (c) replicated from main text Fig. 3b.

abstention recall in Fig. S7, though with underspecified context samples proving more challenging than those with unknown answers or underspecified intent.

D.4 Effect of reasoning fine-tuning

Reasoning inference. In our implementation of inference in reasoning models, we introduce a "forced" reasoning step and a "forced" final answer step. Specifically, the DeepSeek R1 Distill default tokenizer implements chat formatting such that model generations start from a start-of-reasoning token <think>.[1] We implement the same formatting for s1.1, appending its start-of-reasoning tokens <|im_start|>think\n after

[1] See https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-70B for more details.

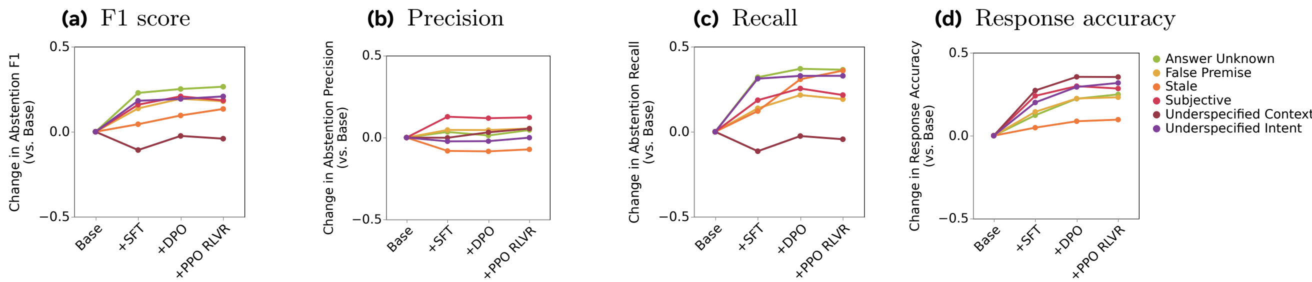

Fig. S5 Change in **(a)** abstention F1 score, **(b)** precision, **(c)** recall, and **(d)** response accuracy of Tülu 8B checkpoints vs. Llama 3.1 base 8B. Panels (c) and (d) replicated from main text Fig. 5.

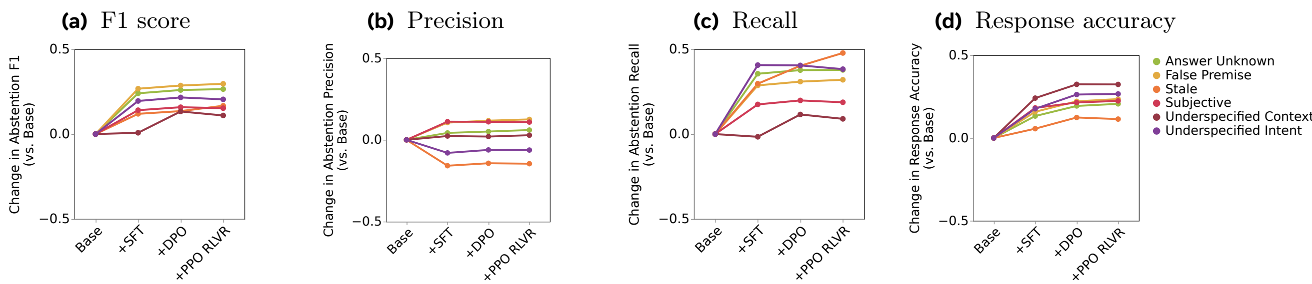

Fig. S6 Change in **(a)** abstention F1 score, **(b)** precision, **(c)** recall, and **(d)** response accuracy of Tülu 70B checkpoints vs. Llama 3.1 base 70B.

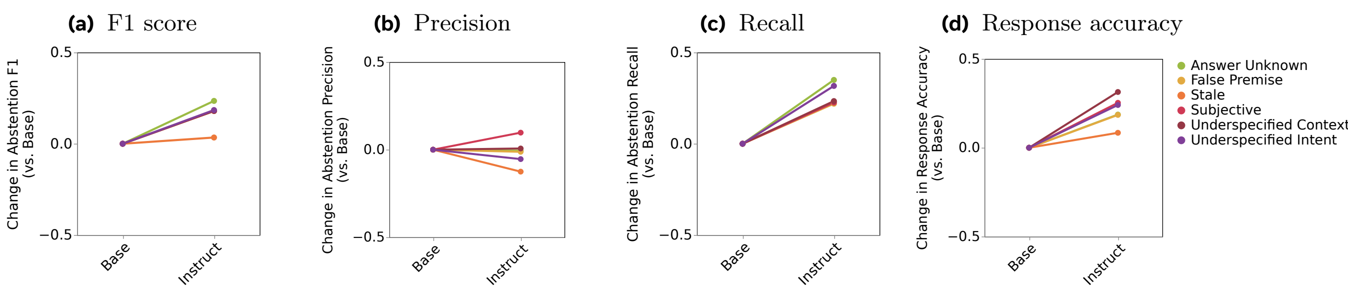

Fig. S7 Change in **(a)** abstention F1 score, **(b)** precision, **(c)** recall, and **(d)** response accuracy of Llama 3.1 8B Instruct vs. Llama 3.1 8B base.

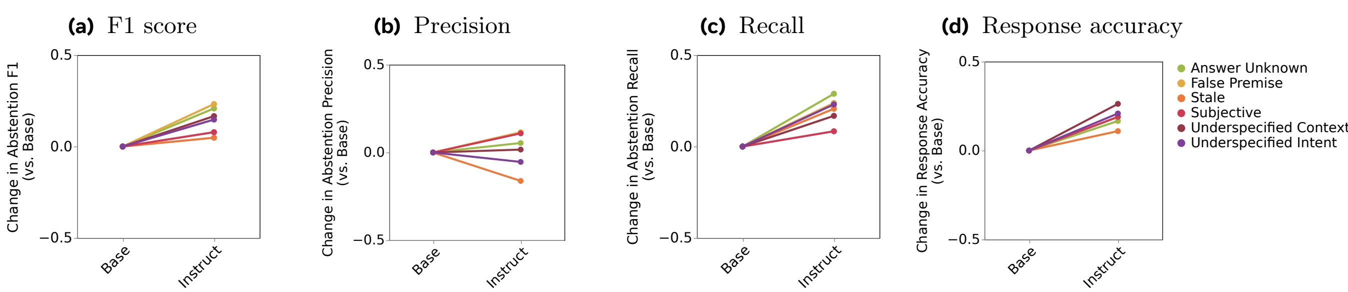

Fig. S8 Change in **(a)** abstention F1 score, **(b)** precision, **(c)** recall, and **(d)** response accuracy of Llama 3.1 70B Instruct vs. Llama 3.1 70B base.